

Eytan Abramowitz

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Education

Columbia University School of Engineering and Applied Science | GPA: 3.89 / 4

Graduation: May 2027

- B.S. Mechanical Engineering | Minors: Aerospace Engineering, Psychology

Relevant Coursework: CFD for Aerospace, Aerodynamics, Dynamics & Vibrations, Mechanics of Solids, Fluid Mechanics, Heat Transfer, Materials & Manufacturing, Computer Science (Java), Data Science (Python), Intro Robotics, Controls, Machine Design

Engineering/Work Experience

Area Maintenance Manager Intern, RME | Amazon

June 2026 - August 2026

- Supervised 16 technicians overseeing 1,000+ work orders across 150+ pieces of equipment and 800+ Kiva robots; led standups to prioritize preventive and reactive tasks across Amazon robotic sortation operations.
- Managed vendor operations, ensuring all personnel met safety standards including LOTO and PPE requirements
- Developed a Python automation tool that integrated 8 internal Amazon data systems to streamline daily KPI reporting across the entire maintenance operation, cutting report generation time by over 90%.

Aerodynamics Team Lead, AIAA Design Build Fly | Columbia University

September 2023 - Present

- Lead a 10-member team responsible for simulation, and manufacturing of aircraft wings and empennage.
- Performed first-principles aerodynamic and structural load calculations to guide preliminary aircraft design.
- Ran CFD simulations to calculate static and dynamic stability for the mission configurations and optimize center-of-gravity placement and weight distribution.
- Benchmarked stability results against military documents, confirming the aircraft met established stability standards.
- Directed fabrication of aerodynamic surfaces using laser cutting, 3D printing, balsa wood, carbon fiber, and Monokote.
- Placed 16th out of 112 teams in the 2024-2025 AIAA DBF competition.

Course Assistant, Mechanics of Solids | Columbia University

September 2025 - December 2025

- Assisted with course administration by grading assignments and answering student questions.

Mechanical Engineer Intern, Creative Machines Lab | Columbia University

June 2025 - December 2025

- Revitalized stalled AI-robot foosball project by designing a compact serving mechanism that improved consistency, reduced hardware footprint by 75%, and reduced cost by 50%.
- Integrated Teknic motors, and wired power distribution system to accelerate the team's testing procedure and ensure electrical safety.
- Developed a rule-based player in C++ using computer vision to benchmark AI agents.

Research Assistant, Meadow Aerodynamics Laboratory | Tel Aviv University

June 2024 - August 2024

- Conducted wind tunnel testing on a novel MEMS bifurcation airflow sensor to validate sensor performance for analyzing flow separation and boundary layer behavior along a Glauert airfoil.
- Utilized Hot-Wire/Film sensors, Pitot tubes, signal generators, voltage multipliers, 3D printing, and Python (Pandas) to collect and analyze data.

Projects

Transonic CFD Analysis of Finite Wing | ANSYS Fluent

- Performed 3D steady-state compressible RANS CFD of finite wing (Mach 0.2–0.89, AoA 0–17°).
- Automated parametric AoA sweeps to compute CL, CD, and CM in body and wind axes, generating aerodynamic characteristic curves for static stability analysis.
- Quantified drag transition to pressure-dominated regime (82% at M=0.89) near transonic conditions.
- Verified boundary layer resolution via y^+ distribution and wall shear stress evaluation.

Aircraft Wing Mount Structural FEA | SolidWorks FEA

- Simulated aircraft wing mounting under flight conditions (10–20 m/s) derived from airfoil lift and drag coefficients.
- Identified worst-case concentrated reactions through first-principles analysis of bounding load cases.
- Redesigned component after determining concentrated loads exceeded PLA tensile strength (~50 MPa), mitigating risk of brittle failure in flight.

Skills

Analysis: ANSYS (CFD, FEA), SolidWorks Simulation, MATLAB, Python (Pandas), LabVIEW

Design & Manufacturing: SolidWorks, Fusion 360, CNC (Mill, Lathe, Laser), 3D Printing

Programming & Control: Java, Arduino, Raspberry Pi, OpenCV, Stable Baselines3 (ML), Motor Integration (Teknic ClearPath)